b. The proof that if N is a normal subgrop and $a, b \in G$, then the subset product $(aN)(bN)$ is contained in the coset $(ab)N$ would just repeat the last paragraph of the proof of Theorem 14.4. To show that $(ab)N$ is contained in $(aN)(bN)$, we let $n \in N$. Then $(ab)n = (ae)(bn)$, and this equation exhibits an element of $(ab)N$ as a product of elements in (aN) and (bN).

c. Associativity was proved for all subsets of G in Part(**a**), so it is surely true for the cosets of the normal subgroup N. Because $N = eN$, we see that $(aN)N = (aN)(eN) = (ae)N = aN$, and similarly $(eN)(aN) = aN$. Thus the coset N acts as identity element. The equation $(a^{-1}N)(aN) = eN = (aN)(a^{-1}N)$ shows that each coset has an inverse, so these cosets of N do form a group under this set multiplication. The identity of the coset group is N, while the identity for the multiplication of all subsets of G is $\{e\}$. If $N \neq \{e\}$, these identities are different.

15. Factor Group Computations and Simple Groups

1. Because $\langle(0,1)\rangle$ has order 4, the factor group has order 2 and must be isomorphic to $\mathbb{Z}_2$. This is also obvious because this factor group essentially collapses everything in the second factor of $\mathbb{Z}_2 \times \mathbb{Z}_4$ to the identity, leaving just the first factor.

2. In this factor group,the first factor is not touched, but in the second factor, the element 2 is collapsed to 0. Because $\mathbb{Z}_4/\langle 2\rangle$ is isomorphic to $\mathbb{Z}_2$, we see the factor group isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$. Alternatively, we can argue that the factor group has order four but no element of order greater than two.

3. We have $\langle(1,2)\rangle = \{(0,0),(1,2)\}$, so the factor group is of order $8/2 = 4$. We easily see that $(1,1) + \langle(1,2)\rangle$ has order 4 in this factor group, which must then be isomorphic to $\mathbb{Z}_4$.

4. We have $\langle(1,2)\rangle = \{(0,0),(1,2),(2,4),(3,6)\}$, so the factor group has order $32/4 = 8$. Because (0, 1) must be added to itself eight times for the sum to lie in $\langle(1,2)\rangle$, we see that $(0,1) + \langle(1,2)\rangle$ is of order 8 in this factor group, which is thus cyclic and isomorphic to $\mathbb{Z}_8$.

5. We have $\langle(1,2,4)\rangle = \{(0,0,0),(1,2,4),(2,0,0),(3,2,4)\}$ so the factor group has order $(4\cdot 4\cdot 8)/4 = 32$. The factor group can have no element of order greater than 8 because $\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_8$ has no elements of order greater than 8. Because (0, 0, 1) must be added to iself eight times for the sum to lie in $\langle(1,2,4)\rangle$, we see that the factor group has an element $(0,0,1) + \langle(1,2,4)\rangle$ of order 8, and is thus either isomorphic to $\mathbb{Z}_4 \times \mathbb{Z}_8$ or to $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_8$. The first group has only three elements of order 2, while the second one has seven elements of order 2. We count the elements of $\mathbb{Z}_4 \times \mathbb{Z}_4 \times \mathbb{Z}_8$ not in $\langle(1,2,4)\rangle$ but which, when added to themselves, yield an element of $\langle(1,2,4)\rangle$. No element added to itself yields (1, 2, 4) or (3, 2, 4). There are six such elements that yield (0, 0, 0) when added to themselves. There are another six such elements that yield (2, 0, 0) when added to themselves. These twelve elements are only enough to form three 4-element cosets in the factor group, which must be ismorphic to $\mathbb{Z}_4 \times \mathbb{Z}_8$.

6. Factoring out by $\langle(0,1)\rangle$ collapses the second factor of $\mathbb{Z} \times \mathbb{Z}$ to zero without touching the first factor, so the factor group is isomorphic to $\mathbb{Z}$. (For those who object to this "collapsing" argument, the projection map $\pi_1 : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z}$ where $\pi_1(m,n) = m$ has $\langle(0,1)\rangle$ as its kernel.)

7. The 1 in the generator (1, 2) of $\langle(1,2)\rangle$ shows that each coset of $\langle(1,2)\rangle$ contains a unique element of the form $(0,m)$, and of course, every such element of $\mathbb{Z} \times \mathbb{Z}$ is in some coset of $\langle(1,2)\rangle$. We can choose these representatives $(0,m)$ to compute in the factor group, which must therefore be isomorphic to $\mathbb{Z}$.

8. The 1 in the generator (1, 1, 1) of $\langle(1,1,1)\rangle$ shows that each coset of $\langle(1,1,1)\rangle$ contains a unique element of the form $(0,m,n)$, and of course, every such element of $\mathbb{Z} \times \mathbb{Z} \times \mathbb{Z}$ is in some coset of

$\langle(1,1,1)\rangle$. We can choose these representatives $(0,m,n)$ to compute in the factor group, which must therefore be isomorphic to $\mathbb{Z}\times\mathbb{Z}$.

9. We conjecture that $(\mathbb{Z}\times\mathbb{Z}\times\mathbb{Z}_4)/\langle(3,0,0)\rangle$ is isomorphic to $\mathbb{Z}_3\times\mathbb{Z}\times\mathbb{Z}_4$, because only the multiples of 3 in the first factor are collapsed to zero. It is easy to check that $\phi:\mathbb{Z}\times\mathbb{Z}\times\mathbb{Z}_4\to\mathbb{Z}_3\times\mathbb{Z}\times\mathbb{Z}_4$ defined by $\phi(n,m,s)=(r,m,s)$, where r is the remainder of n when divided by 3 in the division algorithm, is an onto homomorphism with kernel $\langle(3,0,0)\rangle$. By Theorem 14.6, such a check proves our conjecture.

10. We conjecture that $(\mathbb{Z}\times\mathbb{Z}\times\mathbb{Z}_8)/\langle(0,4,0)\rangle$ is isomorphic to $\mathbb{Z}\times\mathbb{Z}_4\times\mathbb{Z}_8$, because only the multiples of 4 in the second factor are collapsed to zero. It is easy to check that $\phi:\mathbb{Z}\times\mathbb{Z}\times\mathbb{Z}_8\to\mathbb{Z}\times\mathbb{Z}_4\times\mathbb{Z}_8$ defined by $\phi(n,m,s)=(n,r,s)$, where r is the remainder of m when divided by 4 in the division algorithm, is an onto homomorphism with kernel $\langle(0,4,0)\rangle$. By Theorem 14.6, such a check proves our conjecture.

11. Note that $(1,1)+\langle(2,2)\rangle$ is of order 2 in the factor group and $(0,1)+\langle(2,2)\rangle$ generates an infinite cyclic subgroup of the factor group. This suggests that the factor group is isomorphic to $\mathbb{Z}_2\times\mathbb{Z}$. We construct a homomophism ϕ mapping $\mathbb{Z}\times\mathbb{Z}$ onto $\mathbb{Z}_2\times\mathbb{Z}$ having kernel $\langle(2,2)\rangle$. By Theorem 14.6, we will then know that $(\mathbb{Z}\times\mathbb{Z})/\langle(2,2)\rangle$ is isomorphic to $\mathbb{Z}_2\times\mathbb{Z}$.

We want to have $\phi(1,1)=(1,0)$ and $\phi(0,1)=(0,1)$. Because $(m,n)=m(1,1)+(n-m)(0,1)$, we try to define ϕ by

$$\phi(m,n)=(m\cdot 1, n-m).$$

Here $m\cdot 1$ means $1+1+\cdots 1$ for m summands in $\mathbb{Z}_2$, in other words, the remainder of m modulo 2. Because

$$\begin{aligned}\phi[(m,n)+(r,s)] &= \phi(m+r,n+s)\\ &= ((m+r)\cdot 1, n+s-m-r)\\ &= (m\cdot 1, n-m)+(r\cdot 1, s-r)\\ &= \phi(m,n)+\phi(r,s),\end{aligned}$$

we see that ϕ is indeed a homomorphism. For $(r,s)\in\mathbb{Z}_2\times\mathbb{Z}$, we see that $\phi(r,s+r)=(r,s)$, so ϕ is onto $\mathbb{Z}_2\times\mathbb{Z}$. If $\phi(m,n)=(0,0)$, then $m\cdot 1=0$ in $\mathbb{Z}_2$ and $n-m=0$ in $\mathbb{Z}$. Thus m is even and $m=n$, so $(m,n)=(m,m)$ lies in $\langle(2,2)\rangle$. Thus $\mathrm{Ker}(\phi)$ is contained in $\langle(2,2)\rangle$. It is easy to see that $\langle(2,2)\rangle$ is contained in $\mathrm{Ker}(\phi)$, so $\mathrm{Ker}(\phi)=\langle(2,2)\rangle$. As we observed above, Theorem 14.6 shows that our factor group is isomorphic to $\mathbb{Z}_2\times\mathbb{Z}$.

12. Clearly $(1,1,1)+\langle(3,3,3)\rangle$ is of order 3 in the factor group, while $(0,1,0)+\langle(3,3,3)\rangle$ and $(0,0,1)+\langle(3,3,3)\rangle$ both generate infinite subgroups of the factor group. We conjecture that the factor group is isomorphic to $\mathbb{Z}_3\times\mathbb{Z}\times\mathbb{Z}$. As in the solution to Exercise 11, we show that by defining a homomorphism ϕ of $\mathbb{Z}\times\mathbb{Z}\times\mathbb{Z}$ onto $\mathbb{Z}_3\times\mathbb{Z}\times\mathbb{Z}$ having kernel $\langle(3,3,3)\rangle$. Just as in Exercise 11, we are motivated to let

$$\phi(m,n,s)=(m\cdot 1, n-m, s-m).$$

We easily check that ϕ is a homomorphism with the onto property and kernel that we desire, completing the proof. Just follow the arguments in the solution of Exercise 11.

13. Checking the Table 8.12 for D_4, we find that only ρ_0 and ρ_2 commute with every element of D_4. Thus $Z(D_4)=\{\rho_0,\rho_2\}$. It follows that $\{\rho_0,\rho_2\}$ is a normal subgroup of D_4. Now $D_4/Z(D_4)$ has order 4 and is hence abelian. Therefore the commutator subgroup C is contained in $Z(D_4)$. Because D_4 is not abelian, we see that $C\neq\{\rho_0\}$, so $C=Z(D_4)=\{\rho_0,\rho_2\}$.

14. Because ρ_0 is the only element of S_3 that commutes with every element of S_3 (see Table 8.8), we see that $Z(\mathbb{Z}_3 \times S_3) = \mathbb{Z}_3 \times \{\rho_0\}$. Because A_3 is the commutator subgroup of S_3, we see that the commutator subgroup of $\mathbb{Z}_3 \times S_3$ is $\{0\} \times A_3$.

15. From Tables 8.8 and 8.12, $Z(S_3 \times D_4) = \{(\rho_0, \rho_0), (\rho_0, \rho_2)\}$. From Exercise 13, we see that the commutator subgroup is $A_3 \times \{\rho_0, \rho_2\}$.

16. We present the answers in tabular form. The order of the factor group is easy to determine, as we did in Exercises 1 through 12. It is clear the factor groups listed are the only ones possible. Which of the possibilities is the correct one for the given subgroup can easily be determined by taking into account the order, and checking whether there is an element of order 4 in the factor group. We leave the "up to isomorphism" label off the "Factor Group" heading to conserve space.

Subgroup	Factor Group	Subgroup	Factor Group
$\langle(1,0)\rangle$	$\mathbb{Z}_4$	$\langle 2\rangle \times \langle 2\rangle$	$\mathbb{Z}_2 \times \mathbb{Z}_2$
$\langle(0,1)\rangle$	$\mathbb{Z}_4$	$\langle(2,0)\rangle$	$\mathbb{Z}_2 \times \mathbb{Z}_4$
$\langle(1,1)\rangle$	$\mathbb{Z}_4$	$\langle(0,2)\rangle$	$\mathbb{Z}_4 \times \mathbb{Z}_2$
$\langle(1,2)\rangle$	$\mathbb{Z}_4$	$\langle(2,2)\rangle$	$\mathbb{Z}_2 \times \mathbb{Z}_4$
$\langle(2,1)\rangle$	$\mathbb{Z}_4$	$\langle(0,0)\rangle$	$\mathbb{Z}_4 \times \mathbb{Z}_4$
$\langle(1,3)\rangle$	$\mathbb{Z}_4$		

17. The definition is incorrect. Replace "contains" by "consists of".

The **center** of a group G consists of all elements of G that commute with every element of G.

18. The definition is correct.

19. T F F T F T F F T F

20. F/K is isomorphic to $H = \{f \in F \mid f(0) = 0\}$, because every coset of K in F contains a unique function in H, and H is a subgroup of F. There is nothing special about 0 as the choice of a point in the domain of the functions. F/K is also isomorphic to $H_a = \{f \in F \mid f(a) = 0\}$ for the same reason.

21. F^*/K^* is isomorphic to $H^* = \{f \in F^* \mid f(1) = 1\}$, because every coset of K^* in F^* contains a unique function in H^*, and H^* is a subgroup of F^*. There is nothing special about 1 as the choice of a point in the domain of the functions. F^*/K^* is also isomorphic to $H_a^* = \{f \in F^* \mid f(a) = 1\}$ for the same reason.

22. No, if $f + K$ has order 2 in F/K, then we would have to have $f \notin K$ but $g = f + f \in K$. Thus we would have to have g be continuous, but have $f = \frac{1}{2}g$ be not continuous. This is impossible.

23. (See the answer in the text.)

24. U/z_0U is isomorphic to $\{e\}$, for $z_0U = U$.

25. $U/\langle -1\rangle$ is isomorphic to U, for the map $\phi : U \to U$ given by $\phi(z) = z^2$ is a homomorphism of U onto U with kernel $\{-1, 1\}$. By Theorem 14.6, $U/\langle -1\rangle$ is isomorphic to U.

26. $U/\langle \zeta_n\rangle$ is isomorphic to U, for the map $\phi : U \to U$ given by $\phi(z) = z^n$ is a homomorphism of U onto U with kernel $\langle \zeta_n\rangle$. By Theorem 14.6, $U/\langle \zeta_n\rangle$ is isomorphic to U.

27. The factor group $\mathbb{R}/\mathbb{Z}$ is isomorphic to U, the multiplicative group of complex numbers having absolute value 1. The map $\phi : \mathbb{R} \to U$ given by $\phi(r) = e^{(2\pi r)i} = \cos(2\pi r) + i\sin(2\pi r)$ is a homomorphism of $\mathbb{R}$ onto U with kernel $\mathbb{Z}$. By Theorem 14.6, $\mathbb{R}/\mathbb{Z}$ is isomorphic to U.

28. The group $\mathbb{Z}$ is an example, for $\mathbb{Z}/\langle 2\rangle$ has only elements of finite order.

29. Let $G = \mathbb{Z}_2 \times \mathbb{Z}_4$. Then $H = \langle(1,0)\rangle$ is isomorphic to $K = \langle(0,2)\rangle$, but G/H is isomorphic to $\mathbb{Z}_4$ while G/K is isomorphic to $\mathbb{Z}_2 \times \mathbb{Z}_2$.

30. a. The center is a whole group.

b. The center is $\{e\}$, because the center is a normal subgroup and the group is simple.

31. a. The commutator subgroup of an abelian group is {e}.

b. The whole group G is simple. Because C is a normal subgroup, $C = \{e\}$ or $C = G$. Because G is nonabelian and G/C is abelian, we must have $C = G$.

32. Every coset of a factor group G/H that contains a generator of the cyclic group G will generate the factor group.

33. If M and L are normal subgroups of G and $M < L < G$, then L/M is a proper nontrivial normal subgroup of G/M. If $\gamma : G \to G/M$ is the canonical homomorphism and K is a proper nontrivial normal subgroup of G/M, then $\gamma^{-1}[K]$ is a normal subgroup of G and $M < \gamma^{-1}[K] < G$.

34. Every subgroup H of index 2 is normal, because both left and right cosets of H are H itself and $\{g \in G \mid g \notin H\}$. Thus G cannot be simple if it has a subgroup H of index 2.

35. We know that $\phi[N]$ is a subgroup of $\phi[G]$ by Theorem 13.12. We need only show that $\phi[N]$ is normal in $\phi[G]$. Let $g \in G$ and $x \in N$. Because ϕ is a homomophism, Theorem 13.12 tells us that $\phi(g)\phi(x)\phi(g)^{-1} = \phi(g)\phi(x)\phi(g^{-1}) = \phi(gxg^{-1})$. Because N is normal, we know that $gxg^{-1} \in N$, so $\phi(gxg^{-1})$ is in $\phi[N]$, and we are done.

36. We know that $\phi^{-1}[N']$ is a subgroup of G by Theorem 13.12. We need only show that $\phi^{-1}[N']$ is normal in G. Let $x \in \phi^{-1}[N']$, so that $\phi(x) \in N'$. For each $g \in G$, we have $\phi(gxg^{-1}) = \phi(g)\phi(x)\phi(g^{-1}) = \phi(g)\phi(x)\phi(g)^{-1} \in N'$ because N' is a normal subgroup of G'. Thus $gxg^{-1} \in \phi^{-1}[N']$, showing that $\phi^{-1}[N']$ is a normal subgroup of G. (Note that N' need only be normal in $\phi[G]$ for the conclusion to hold.)

37. Suppose that $G/Z(G)$ is cyclic and is generated by the coset $aZ(G)$. Let $x, y \in G$. Then x is a member of a coset $a^m Z(G)$ and y is a member of a coset $a^n Z(G)$ for some $m, n \in \mathbb{Z}$. We can thus write $x = a^m z_1$ and $y = a^n z_2$ where $z_1, z_2 \in Z(G)$. Because z_1 and z_2 commute with every element of G, we have $xy = a^m z_1 a^n z_2 = a^{m+n} z_1 z_2 = a^n z_2 a^m z_1 = yx$, showing that G is abelian. Therefore, if G is not abelian, then $G/Z(G)$ is not cyclic.

38. Let G be nonabelian of order pq. Suppose that $Z(G) \neq \{e\}$. Then $|Z(G)|$ is a divisor of pq greater than 1, but less than pq because G is nonabelian, and hence $|Z(G)|$ is either p or q. But then $|G/Z(G)|$ is either q or p, and hence is cyclic, which contradicts Exercise 37. Therefore $Z(G) = \{e\}$.

39. a. Because $(a,b,c) = (a,c)(a,b)$, we see that every 3-cycle is an even permutation, and hence is in A_n.

b. Let $\sigma \in A_n$ and write σ as a product of transpositions. The number of transpositions in the product will be even by definition of A_n. The product of the first two transpositions will be either of the form $(a,b)(c,d)$ or of the form $(a,b)(a,c)$ or of the form $(a,b)(a,b)$, depending on repetition of letters in the transpositions. If the form is $(a,b)(a,b)$, it can be deleted from the product altogether. As the hint shows, either of the other two forms can be expressed as a 3-cycle. We then proceed with

the next pair of transpositions in the product, and continue until we have expressed σ as a product of 3-cycles. Thus the 3-cycles generate A_n.

c. Following the hint, we find that

$$\begin{gathered}(r,s,i)^2 = (r,i,s),\\ (r,s,j)(r,s,i)^2 = (r,s,j)(r,i,s) = (r,i,j),\\ (r,s,j)^2(r,s,i) = (r,j,s)(r,s,i) = (s,i,j),\\ (r,s,i)^2(r,s,k)(r,s,j)^2(r,s,i) = (r,i,s)(r,s,k)(s,i,j) = (i,j,k).\end{gathered}$$

Now every 3-cycle either contains neither r nor s and is of the form (i,j,k), or just one of r or s and is of the form (r,i,j) or (s,i,j), or both r and s and is of the form (r,s,i) or $(r,i,s) = (s,r,i)$. Beause all of these forms can be obtained from our special 3-cycles, we see that the special 3-cycles generate A_n.

d. Following the hint and using Part(**c**) we find that

$$((r,s)(i,j))(r,s,i)^2((r,s)(i,j))^{-1} = (r,s)(i,j)(r,i,s)(i,j)(r,s) = (r,s,j).$$

Thus if N is a normal subgroup of A_n and contains a 3-cycle, which we can consider to be (r,s,i) because r and s could be any two numbers from 1 to n in Part(**c**), we see that N must contain all the special 3-cycles and hence be all of A_n by Part(**c**).

e. Before making the computations in the hints of the five cases, we observe that one of the cases must hold. If Case 1 is not true and Case 2 is not true, then when elements of N are written as a product of disjoint cycles, no cycle of length greater than 3 occurs, and no element of N is a single 3-cycle. The remaining cases cover the possibilities that at least one of the products of disjoint cycles involves two cycles of length 3, involves one cycle of length 3, or involves no cycle of length 3. Thus all possibilities are covered, and we now turn to the computations in the hints.

Case 1. By Part(**d**), if N contains a 3-cycle, then $N = A_n$ and we are done.

Case 2. Note that $a_1, a_2, \cdots, a_r$ do not appear in μ because the product contained disjoint cycles. We have

$$\begin{gathered}\sigma^{-1}[(a_1,a_2,a_3)\sigma(a_1,a_2,a_3)^{-1}]\\ = (a_r,\cdots,a_2,a_1)\mu^{-1}(a_1,a_2,a_3)\mu(a_1,a_2,\cdots,a_r)(a_1,a_3,a_2)\\ = (a_1,a_3,a_r),\end{gathered}$$

and this element is in N because it is the product of σ^{-1} and a conjugate of σ by an element of A_n. Thus in this case, N contains a 3-cycle and is equal to A_n by Part(**d**).

Case 3. Note that $a_1, a_2, \cdots, a_6$ do not appear in μ. As in Case 2, we see that

$$\begin{gathered}\sigma^{-1}[(a_1,a_2,a_4)\sigma(a_1,a_2,a_4)^{-1}] =\\ (a_1,a_3,a_2)(a_4,a_6,a_5)\mu^{-1}(a_1,a_2,a_4)\mu(a_4,a_5,a_6)(a_1,a_2,a_3)(a_1,a_4,a_2)\\ = (a_1,a_4,a_2,a_6,a_3)\end{gathered}$$

is in N. Thus N contains a cycle of length greater than 3, and $N = A_n$ by Case 2.

Case 4. Note that a_1, a_2, and a_3 do not appear in μ. Of course $\sigma^2 \in N$ because $\sigma \in N$, so $\sigma^2 = \mu(a_1,a_2,a_3)\mu(a_1,a_2,a_3) = (a_1,a_3,a_2) \in N$, so N contains a 3-cycle and hence $N = A_n$ as shown by Part(**d**).

Case 5. Note that a_1, a_2, a_3, and a_4 do not appear in μ. As in Case 2, we see that